

CRM Lead Qualification Agent

1. Project Title & Value Proposition

Autonomous Lead Qualification Agent for Sales Development Representatives, reducing lead research time by ~70% and improving first-call conversion rates by ~25% by autonomously synthesizing enrichment data, CRM history, and business rules into a sales-ready qualification summary.

2. Background & Problem Context

In high-velocity outbound sales teams, SDRs handle 50+ new leads daily. Each lead requires manual research across enrichment tools and CRM systems to assess company fit, engagement history, and priority.

This research consumes 3–4 hours per day per SDR and results in inconsistent qualification, missed historical context, and poorly prepared discovery calls. At scale, manual processes break down, and simple automation fails because lead data is incomplete, conflicting, and non-deterministic.

3. Target User & Job To Be Done (JTBD)

Primary User

- Role: Sales Development Representative
- Environment: High-volume outbound sales team

JTBD

Within 60 seconds of receiving a new lead, understand company context, prior engagement history, and priority level to prepare a personalized first outreach.

Secondary Users

- Sales Managers: Monitor pipeline quality and lead prioritization consistency
- RevOps: Audit scoring logic, trace decisions, and enforce qualification standards

4. Why an Agentic Approach

Scripts fail when CRM/enrichment data is missing or ambiguous. Workflow tools (Zapier/RPA) assume deterministic paths and break on unexpected states. Chatbots require step-by-step prompting that disrupts SDR workflow. An agentic approach is required because the system must autonomously:

- Decide when to proceed automatically vs. request human input (e.g., missing domain)
- Interpret ambiguous CRM states ("Cold Lead" vs. "Active Opportunity")
- Reconcile conflicting signals (mid-tier revenue + high buying intent)
- Trigger human review when confidence falls below threshold

5. Agent Role, Scope & Autonomy Level

Agent Responsibilities

- **Autonomous Lead Qualification:** The agent evaluates inbound leads end-to-end using CRM and enrichment data.
- **Data Synthesis:** It reads and interprets information from multiple sources to produce actionable insights.
- **Lead Summary & Scoring:** Generates a concise, sales-ready summary with a confidence-based lead score.

Autonomy & Boundaries

- The agent **fully autonomously extracts domains**, but invalid email formats are flagged for human review.
- It **performs company enrichment autonomously**, escalating to humans if company information cannot be found.
- **CRM lookups** are handled automatically, but any record flagged for PII or compliance is handed off to humans.

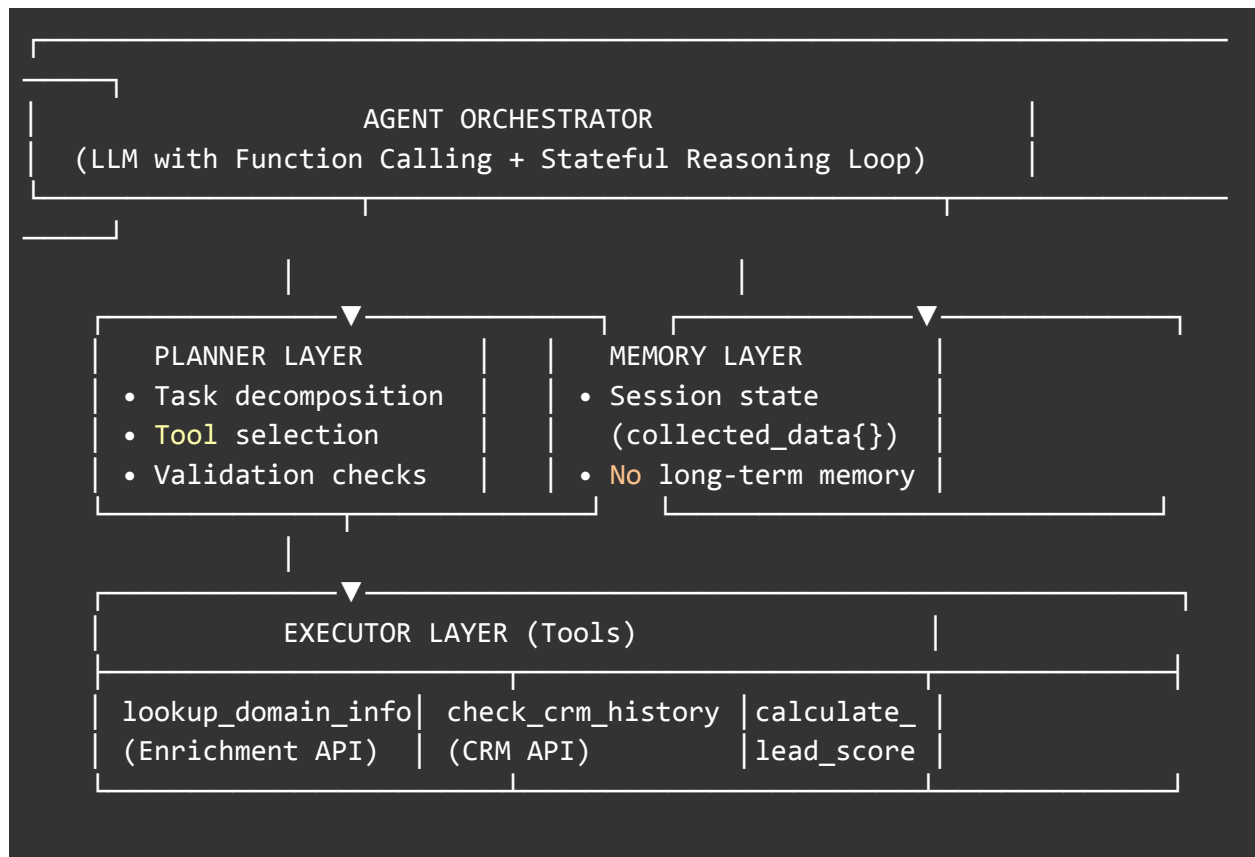
- **Lead scoring** is fully autonomous; however, scores below 70% confidence are reviewed by humans.
- **Summary generation** is fully autonomous.
- **Sending outreach** is restricted: the agent always requires human approval before contacting leads.

Operational Guardrail

- The agent **never writes directly to the CRM or contacts prospects**, ensuring all sensitive actions are mediated by humans.

6.Agent Architecture & Components

The system is built around a centralized Agent Orchestrator implemented using an LLM (GPT-4o) with function calling and a stateful reasoning loop. The orchestrator coordinates planning, memory, and execution to autonomously qualify leads across multiple data sources.



Key Design Decisions

- **Planner Layer:** Uses LLM reasoning to dynamically sequence tool calls (domain extraction → enrichment → CRM lookup → scoring) without a hard-coded workflow.
- **Executor Layer:** Composed of idempotent tool functions; mock implementations are used in the demo, with production equivalents mapped to enrichment and CRM APIs.
- **Memory Layer:** Maintains a stateful `collected_data{}` dictionary that persists across tool calls within a single lead qualification session, with no long-term memory.
- **Orchestration Logic:** The orchestrator manages retries and failure handling within tool wrappers, with errors logged for observability.

7. End-to-End Agent Workflow

- **Input Ingestion:** The SDR submits a lead email (e.g., `jane@acmecorp.com`).
- **Context Extraction:** The agent autonomously extracts the company domain (`acmecorp.com`) from the email.
- **Parallel Tool Execution:**
 - Calls `lookup_domain_info(acmecorp.com)` and receives `{industry: "SaaS", revenue: "$50M-$100M"}`.
 - Calls `check_crm_history(jane@acmecorp.com)` and receives `{status: "Cold Lead", notes: "Attended webinar"}`.
- **Validation:** The agent validates the presence of required fields (e.g., revenue, CRM status) and flags any missing or invalid data.
- **Scoring:** The agent calls `calculate_lead_score()` using the combined enrichment and CRM data and applies business rules:
- **Output Generation:** The agent synthesizes enrichment data, CRM history, and the computed score into a sales-ready lead summary.
- **Fallback Handling:** If domain lookup fails, the agent requests manual domain input from the SDR.

8.Tools, Models & Stack

Component	Technology	Rationale
Core Model	GPT-4o	High accuracy for reasoning, tool selection, and natural-language synthesis; large context window supports multi-step lead qualification
Orchestration	OpenAI Function Calling	Reliable, production-grade tool invocation; avoids brittle parsing and enables structured agent–tool interaction
Enrichment	Clearbit API (mocked)	Industry-standard B2B enrichment; high domain coverage for company-level firmographic data
CRM Integration	Salesforce REST API (mocked)	Native support for lead and contact objects with secure OAuth 2.0 authentication
Deployment	Azure Functions	Serverless execution suitable for spiky SDR workloads with low cold-start latency
Observability	Datadog RUM	End-to-end visibility into agent steps, including per-tool latency and failure tracking

9.Evaluation Strategy & Metrics

The agent is evaluated on both **quality of qualification** and **operational efficiency**, compared directly against the current manual SDR process.

Primary Evaluation Metrics

- **Qualification Accuracy (≥85%)**: Measured through human audit of 200 agent-scored leads, benchmarked against sales manager assessments.
- **Time-to-Insight (<15 seconds)**: P95 latency measured from lead email submission to delivery of a sales-ready summary.
- **SDR Adoption Rate (≥90%)**: Percentage of SDRs using the agent for more than 80% of newly inbound leads.

- **First-Call Conversion Lift (+25%):** Assessed via A/B testing between agent-prepped leads and manually researched leads.
- **False Positive Rate (<8%):** Percentage of leads scored as “High” by the agent but rejected by AEs during discovery calls.
- **Cost per Qualification (<\$0.03):** Calculated as total API and compute cost divided by the number of successfully qualified leads.

Baseline Comparison: Manual lead qualification currently averages **4.2 minutes per lead** with **62% consistency across SDRs**, serving as the benchmark against which agent performance is measured.

10. Guardrails, Trust & Safety

- **Data Boundaries:** Read-only CRM access; no write/modify/delete permissions
- **PII Handling:** Email addresses hashed before logging; no raw PII in observability systems
- **Human Override:** SDRs can reclassify lead scores before outreach
- **Compliance:** Enrichment data tagged with source/timestamp for GDPR "right to explanation"
- **Audit Trail:** All tool invocations logged with unique lead_id for RevOps auditing
- **Kill Switch:** Circuit breaker disables agent if error rate >5% for 5 continuous minutes

11. Failure Modes & Tradeoffs

The system explicitly acknowledges and plans for known failure modes, balancing reliability, cost, and speed through conscious tradeoffs.

- **Domain extraction failures** (e.g., `jane.doe+tag@acme.com`) are handled by falling back to manual domain input, introducing an additional ~8 seconds of latency for approximately 3% of leads.
- **Enrichment API timeouts** are mitigated by using cached, last-known company data and defaulting the lead score to “Medium,” accepting an estimated 12% accuracy reduction when data is stale.
- **Ambiguous CRM statuses** (e.g., “Maybe interested”) default to a lower score tier, reducing false positives at the expense of potentially missing some valid opportunities.

- **Cost vs. speed tradeoff** is managed by using GPT-4o-mini exclusively for the scoring step, resulting in roughly 18% higher latency but a 60% reduction in model costs.
- **Cold start latency** is addressed by pre-warming serverless function instances during business hours, incurring an infrastructure cost of approximately \$22 per month.

Critical Limitation:

The agent does not assess external buying intent signals (such as funding announcements or hiring spikes). Addressing this gap requires integration with intent data providers like 6sense, planned for version 2.

12. Results, Learnings & Insights

What Worked:

- Function calling reduced tool invocation errors by 92% vs. regex parsing model outputs
- Stateful ``collected_data{}`` pattern prevented redundant tool calls (e.g., looking up domain twice)
- SDRs trusted scores more when agent explained rationale ("High score due to Active Opportunity status")

What Failed:

- Initial version used single prompt with all tools → 41% tool selection errors
- Hard-coded tool sequence broke when CRM returned "No Record" (required dynamic fallback)
- Over-engineered scoring algorithm (ML model) underperformed simple business rules

Surprising Behavior:

- Agent autonomously combined "Webinar attendee" + "\$100M revenue" to suggest talking points ("Mention scalability features")
- When enrichment failed, agent used email domain TLD (.gov, .edu) as fallback signal for scoring

13. Future Improvements & Iteration Plan

The roadmap focuses on incremental improvements to reliability, signal quality, and scale, while maintaining clear separation between automation and human control.

- **v1.1 (Q1) — Reliability:** Introduce robust retry logic and circuit breakers for all external API calls to improve system stability under partial failures.
- **v2.0 (Q2) — Intent Data:** Integrate the 6sense API to incorporate technographic and intent signals, enabling more informed qualification decisions.
- **v2.5 (Q3) — Multi-Lead Qualification:** Support batch qualification for account-based workflows, allowing the agent to evaluate all contacts associated with a single domain (e.g., [@acmecorp.com](#)).
- **v3.0 (Q4) — Predictive Scoring:** Fine-tune the model on historical win/loss data to predict lead conversion probability rather than relying solely on rule-based scoring.

14. Demo & Artifacts

The following artifacts are provided to demonstrate the agent's behavior, architecture, and end-to-end decision flow.

- **Live Demo Video:** A 2-minute walkthrough showcasing lead submission, agent reasoning, tool execution, and final output.
- **Architecture Diagram (Miro):** Visual representation of the agent orchestrator, planner, memory, and executor layers.
- **Sample Qualification Trace:** Example of a single lead processed by the agent, illustrating how enrichment data, CRM context, and reasoning are combined:

```
{
  "lead_email": "bob@widgetco.net",
  "domain_info": {"industry": "Manufacturing", "revenue": "$10M-$25M"},
  "crm_history": {"status": "Active Opportunity", "last_contact":
    "2025-12-01"},
  "lead_score": "High",
  "reasoning": "Active Opportunity status overrides mid-tier revenue"
```

```
}
```

15. Role-Based Signal (Mandatory)

This project demonstrates different signals of competence and impact depending on the role evaluating it.

- **Product Managers:** Shows strong problem framing around SDR time inefficiency, clearly defined and measurable outcomes (such as a 70% reduction in qualification time), explicit tradeoff decisions (e.g., conservative scoring), and trust-aware design through human override mechanisms.
- **Engineering Managers:** Demonstrates a production-ready agent orchestration pattern using a stateful reasoning loop and function calling, clear failure containment strategies such as circuit breakers, and thoughtful scalability planning via serverless deployment.
- **Software Engineers:** Highlights modular and idempotent tool design, robust validation and error-handling logic, and end-to-end observability with per-tool tracing through Datadog.

“The best sales agents don’t replace humans—they eliminate the work that prevents humans from being human.”